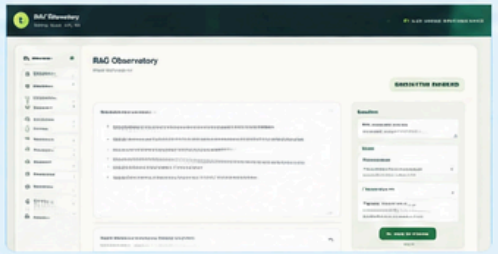
RAG EVAL 2.0 – Current System Architecture

Multimodal Document Ingestion, Retrieval-Augmented Generation, and Evaluation

User Interface (Web Frontend)

PHP + HTML/CSS/JavaScript



Chat
- Ask questions
- Configure retrieval/generation
- View retrieved sources

Documents
- Upload and manage sources
- TXT, PDF, DOCX, PNG, JPG, JPEG
- Category and type filtering

Gold Standard
- Manage evaluation dataset
- View and edit questions

Evaluation
- Run and review evaluations
- Compare results and metrics

Configuration
- Model settings
- Retrieval parameters
- Cost controls and environment flags

HTTP Requests (REST API)

JSON Responses (answers, sources, evaluations, etc.)

Backend API (PHP)

REST Endpoints

/api/ask.php
Handle chat requests and return grounded answers

/api/documents.php
Upload, list, and manage documents

/api/evaluations.php
Run evaluations and retrieve results

/api/health.php
System health check

Query Request (question + settings)

Answer + Sources + Metadata

Upload Request (file + metadata)

Ingestion Status (document info)

Document Ingestion Pipeline (Python)

Supported Formats: TXT, PDF, DOCX, PNG, JPG, JPEG

1. File Validation
Check format, size, and readability

2. Content Extraction
- Text extraction (PDF, DOCX, TXT)
- Table extraction (PDF)
- Image analysis (PDF embedded images, charts, diagrams, and standalone PNG/JPG)

3. Chunk Creation
Split into searchable chunks with metadata (document, page, element type, source, etc.)

4. Embedding Generation
Create vector embeddings for each chunk

5. Store in Databases
Save document records and chunk metadata

Image/Visual Content for Analysis

Textual Description of Visual Content (searchable)

Vision Language Model (Visual Content Analysis)

Google Gemini
(API – cloud)
e.g., gemini-3.1-flash-lite

or

Ollama
(local)
e.g., Qwen2.5-VL 7B

Retrieval System (Python)

Hybrid Search (configurable)

ChromaDB (Vector Search)
- Vector embeddings
- Semantic similarity
- Returns top-k chunks

Combined Results

MySQL (Keyword Search)
- Keyword matching
- Title/filename/metadata
- Returns top-k chunks

Store document records and chunk embeddings

MySQL Database
- Document records
- Chunk metadata
- Evaluation data
- Application state

ChromaDB Vector Database
- Chunk embeddings
- Vector index
- Similarity search

Top-k retrieved chunks (relevant context)

Answer Generation (Python)

LLM with Retrieved Context

Prompt Construction
- User question
- Retrieved chunks (top-k)
- Instructions for grounded answer

Google Gemini
(API – cloud)
e.g., gemini-3.6-flash or gemini-3.1-flash-lite

Grounded Answer with Citations and Metadata

Saved responses (for evaluation)

Evaluation Framework (Python)

Local Evaluators (RAGAS / LLM-judges)

Evaluation Pipeline
- Run saved questions
- Generate answers
- Apply local evaluators
- Store results in database

Evaluation Metrics
- Relevance
- Faithfulness
- Context precision/recall
- Answer correctness
- Safety and other policies

Evaluation results and scores